



UNIVERSITY COLLEGE OF ENGINEERING KANCHEEPURAM
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COMPUTER SCIENCE AND ENGINEERING

**Agentic AI for Personal Task Planning and
Schedule Optimization**

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INTRODUCTION

- ❖ **PROBLEM STATEMENT:** Today, people manage many daily tasks such as meetings, reminders, emails, and schedules using multiple apps. Managing everything manually is time-consuming and error-prone. Existing systems do not intelligently coordinate tasks or provide smart suggestions.
- ❖ **PROPOSED SOLUTION:** This project proposes an **AI-based multi-agent system** where multiple intelligent agents work together to understand user tasks, plan activities, execute actions, and verify results automatically.
- ❖ **KEY TECHNOLOGIES USED:** Large Language Models (LLMs), Multi-Agent Systems, Task Planning and Coordination and Intelligent Automation.

ABSTRACT

1. Recent advancements in Large Language Models have made it possible for AI systems to understand natural language and assist users in real-life tasks.
2. This project focuses on building a **cooperative multi-agent system** where different agents work together to automate personal tasks such as scheduling, reminders, and planning activities.
3. Each agent has a specific role, such as understanding user requests, planning tasks, executing actions, and validating outputs. By working together, the system provides smarter and more reliable task automation.
4. The proposed system aims to reduce human effort, improve productivity, and demonstrate how agent-based AI can be applied to everyday personal task management.

AIM & OBJECTIVE

AIM : To design and develop an AI-powered cooperative multi-agent system for automating personal daily tasks.

OBJECTIVES :

- ❖ To build multiple agents that work together to handle tasks
- ❖ To automate task analysis, planning, validation and execution
- ❖ To reduce manual task management for users
- ❖ To improve task accuracy using agent collaboration

SCOPE

- ❖ Can be used for personal task management such as scheduling and reminders
- ❖ Suitable for students, working professionals, and individuals
- ❖ Can be extended to email automation, calendar management, and reminders
- ❖ Supports future improvements with advanced AI models
- ❖ Acts as a foundation for intelligent personal assistants
- ❖ **IMPACT:** The system reduces manual effort, improves task accuracy, and helps users manage daily activities efficiently using AI assistance.

EXISTING SYSTEM

- ❖ Users manually manage tasks using calendars and reminder apps.
- ❖ Traditional task management apps are rule-based
- ❖ Single AI systems handle tasks independently
- ❖ No intelligent coordination between multiple actions
- ❖ Limited automation and decision-making capability

EXISTING SYSTEM -DRAWBACKS

- ❖ **Lack of Intelligence:** No smart understanding of user intent
- ❖ **Manual Effort:** Users must enter and manage tasks manually
- ❖ **No Validation:** Errors are not automatically detected
- ❖ **Poor Coordination:** Tasks are not connected or optimized
- ❖ **Low Automation:** Requires frequent user interaction

LITERATURE REVIEW

S.NO	TITLE	AUTHORS & YEAR	TECHNIQUE USED	DRAWBACKS
1	LLM-Based Agent Systems for Task Automation	JinlongWang , YuntingWu 2025	AI-driven task planning and execution	Mostly single-agent systems
2	Multi-Agent Reinforcement Learning for Task Offloading in Crowd-Edge Computing	Su Yao, Mu Wang, Ju Ren - 2025	Rule-based and AI-based automation	Limited adaptability

LITERATURE REVIEW

S.NO	TITLE	AUTHORS & YEAR	TECHNIQUE USED	DRAWBACKS
3	Multi-Agent Systems for Complex Problem Solving	Wooldridge, 2024	Agent collaboration and coordination frameworks	High coordination and communication overhead
4	AI-Based Task Scheduling	Kumar et al., 2023	Automated scheduling algorithms	Limited reasoning capability

PROPOSED SYSTEM

System Components:

- ❖ **User Interaction Agent:** Understands user requests
- ❖ **Planning Agent:** Breaks tasks into steps
- ❖ **Execution Agent:** Performs assigned actions
- ❖ **Validation Agent:** Checks correctness of results
- ❖ **Coordinator Agent:** Manages communication between agents

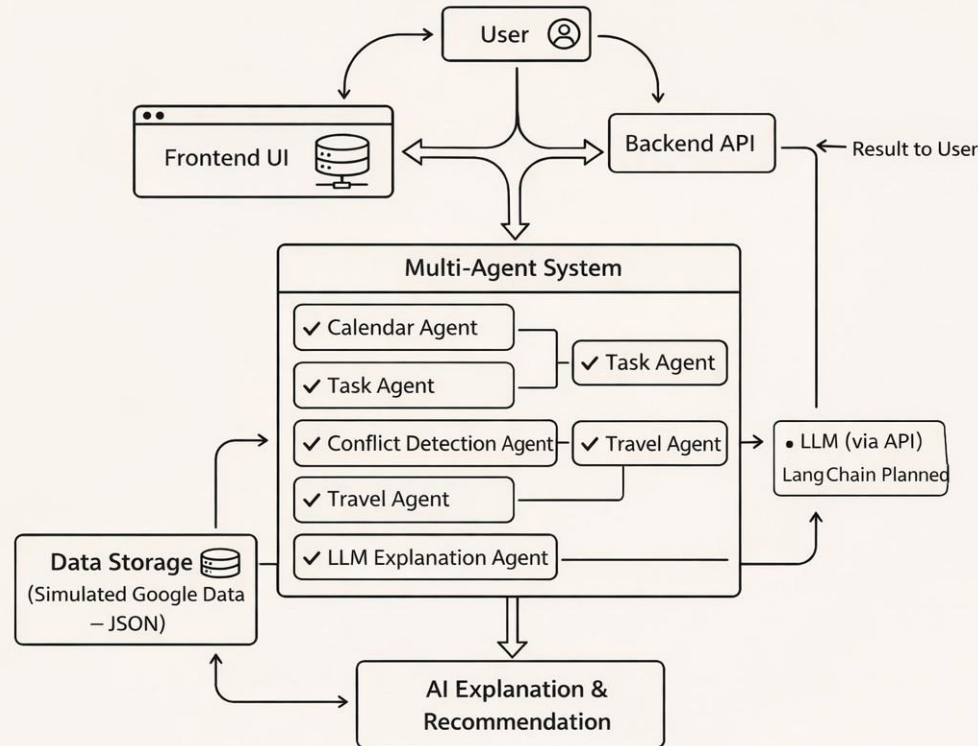
Working Principle:

- ❖ Agents communicate with each other, share information, validate outputs, and complete tasks collaboratively.

PROPOSED SYSTEM ADVANTAGES

- ❖ Improved task accuracy
- ❖ Reduced manual effort
- ❖ Better task coordination
- ❖ Intelligent decision-making
- ❖ Scalable and flexible design
- ❖ User-friendly automation

ARCHITECTURAL DIAGRAM



MODULES

❖ **User Authentication Module**

Handles user **registration, login, and Google OAuth authentication**, ensuring secure access to the system

❖ **Task & Project Management Module**

Allows users to **create, update, prioritize, and organize tasks into projects** for efficient productivity management

❖ **AI Chat Assistant Module**

Provides a **natural language interface** where users can interact with the system to generate tasks and get productivity assistance

MODULES

❖ Voice Assistant Module

Enables **hands-free interaction using voice commands**, including a main voice interface and a floating voice assistant widget.

❖ Chat History Management Module

Stores and manages **conversation history between the user and AI assistant** to maintain interaction context.

❖ Backend & Database Module

Implements system logic using **FastAPI with PostgreSQL database** for storing users, tasks, projects, and chat sessions.

METHODOLOGY

1. User Interaction & Request Input

- ❑ Users interact with the system through **chat, voice commands, or manual task input**.
- ❑ The system captures user requests and converts them into structured task information.

2. Authentication & User Verification

- ❑ The system verifies users through **secure login or Google OAuth authentication**.
- ❑ User identity and session data are maintained for personalized task management.

3. Task Processing & Organization

- ❑ User requests are processed to **create, update, or manage tasks and projects**.
- ❑ Tasks are stored with attributes such as **priority, status, and deadlines**.

4. AI Assistance & Task Generation

- ❑ The AI assistant analyzes user prompts using **LLM-based reasoning**.
- ❑ It provides **task suggestions, productivity guidance, and contextual responses**.

METHODOLOGY

5. Voice Command Processing

- ❑ Voice inputs are converted into **text commands using speech recognition**.
- ❑ The system interprets commands to **perform task creation or system actions**.

6. Data Storage and Management

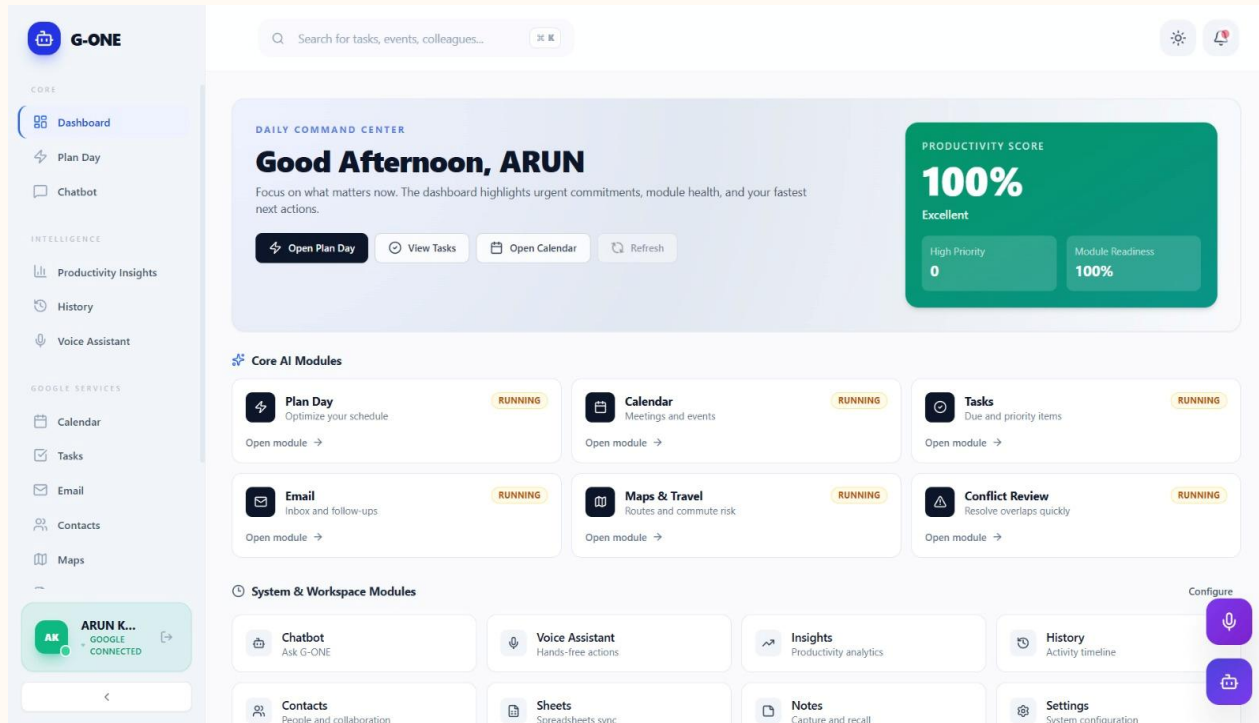
- ❑ All user data including **tasks, projects, and chat sessions** are stored in a PostgreSQL database.
- ❑ Backend APIs ensure **secure and efficient data operations**.

7. Response Generation & User Feedback

- ❑ The system generates **AI responses, task updates, or confirmations**.
- ❑ Results are returned to the user through the **chat interface or voice assistant**.

PROPOSED WORK

DASHBOARD OF G-ONE



PROPOSED WORK

PLAN A DAY BY G-ONE

The screenshot displays the G-ONE AI Day Planner interface. On the left is a sidebar with navigation options: CORE (Dashboard, Plan Day, Chatbot), INTELLIGENCE (Productivity Insights, History, Voice Assistant), and GOOGLE SERVICES (Calendar, Tasks, Email, Contacts, Maps, Sheets). The main panel is titled "AI Day Planner" and "Multi-Agent Orchestration Engine". It features a search bar at the top and a "To exit full screen, press and hold Esc" overlay. The interface is divided into several sections: PLAN SETTINGS (DATE: 17-03-2026, TIMEZONE: Asia/Calcutta, WORK START: 09:00, WORK END: 18:00), PRODUCTIVITY STRATEGY (Balanced, Aggressive, Relaxed), MEETINGS (0), and TASKS (2). The TASKS section shows a task for "client" with a duration of 30 minutes, priority of Medium, and category of Work. On the right, there are four status cards: CONFLICTS (0), TRAVEL TIME (0m), PRODUCTIVITY (100%), and OVERLOAD RISK (Low). Below these is a "Generated at 11:37:59 AM" timestamp and buttons for "Re-optimize" and "Export JSON". The bottom right section, titled "AI Summary", provides a "Good morning! Let's plan your day." message, a "Your Schedule Overview" (a free day with no meetings), "Key Points to Note" (two medium-priority tasks, no conflicts, free day), "Here's What I Recommend" (start day with tasks, take breaks, use free time for deep work), and "Travel Considerations" (no travel needed).

G-ONE

CORE

- Dashboard
- Plan Day
- Chatbot

INTELLIGENCE

- Productivity Insights
- History
- Voice Assistant

GOOGLE SERVICES

- Calendar
- Tasks
- Email
- Contacts
- Maps
- Sheets

AK ARUN K... CHECKING...

Search for tasks, events, colleagues...

To exit full screen, press and hold Esc

AI Day Planner

Multi-Agent Orchestration Engine

Manual Live (Google) DONE Generate Optimized Plan

PLAN SETTINGS

TIME SETTINGS

DATE: 17-03-2026 TIMEZONE: Asia/Calcutta

WORK START: 09:00 WORK END: 18:00

PRODUCTIVITY STRATEGY

Balanced Aggressive Relaxed

Mix of focus & flexibility Pack schedule tightly Generous buffers

Auto-fill from Google Sample Scenario

MEETINGS 0

+ Add Meeting

TASKS 2

client

Duration (min): 30 Priority: Medium Category: Work

Travel Flexible

CONFLICTS 0

TRAVEL TIME 0m

PRODUCTIVITY 100%

OVERLOAD RISK Low

Generated at 11:37:59 AM

Re-optimize Export JSON

Schedule Conflicts Travel Plan AI Explanation Agent Data

AI Summary

Good morning! Let's plan your day.

Your Schedule Overview:

You have a free day with no meetings scheduled, which is great for deep work. You have two tasks to focus on, both related to clients, with a moderate workload assessment.

Key Points to Note:

- You have two medium-priority tasks to complete, each with a duration of 30 minutes.
- There are no conflicts or scheduling issues to worry about.
- Your entire day is free, allowing you to focus on your tasks without any interruptions.

Here's What I Recommend:

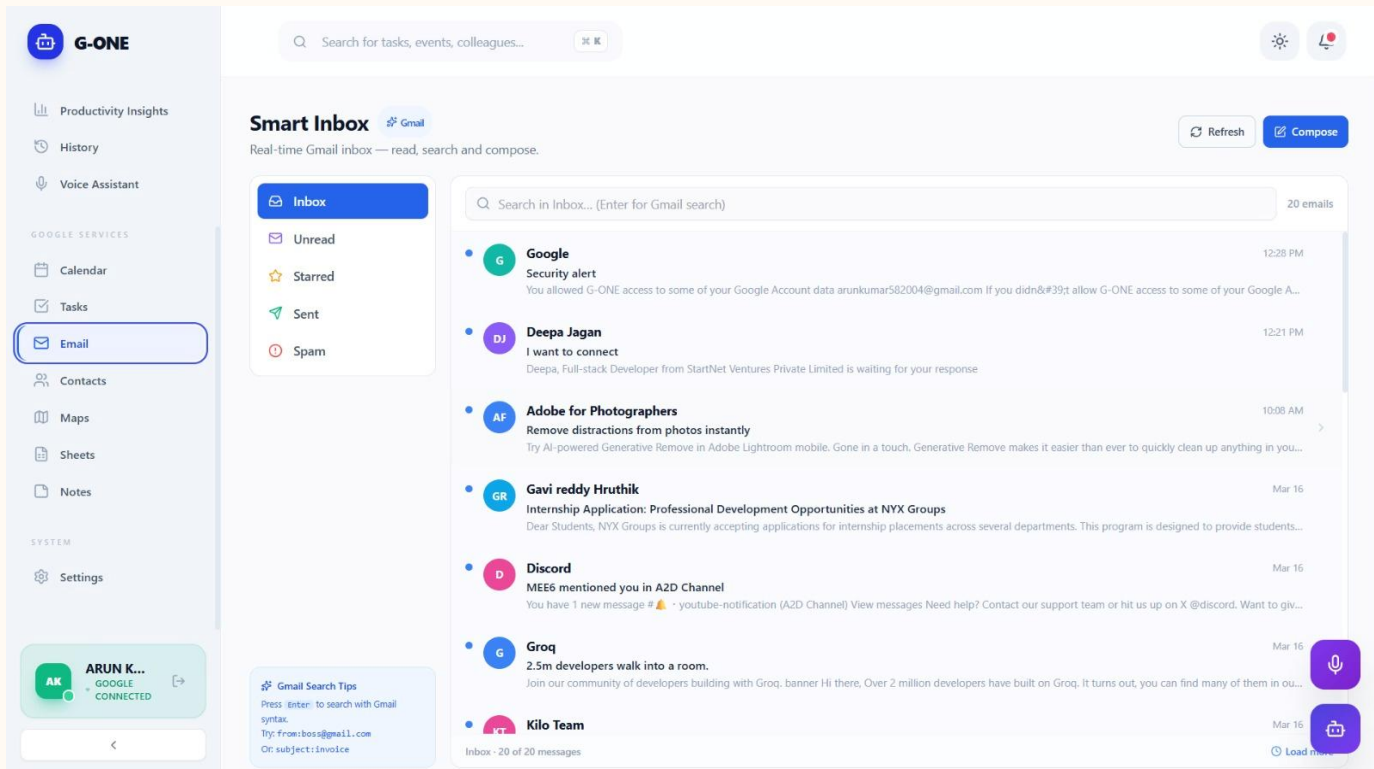
- * Start your day by tackling your two client tasks. Since they are of medium priority, try to complete them as efficiently as possible.
- * Take regular breaks to stay focused and refreshed throughout the day.
- * Use your free time to dive into deep work, making the most of your uninterrupted schedule.

Travel Considerations:

No travel is needed today, so you can stay focused on your tasks without worrying about commuting.

PROPOSED WORK

EMAIL



PROPOSED WORK

CHATBOT

The screenshot displays the G-One Assistant interface, which is a multi-agent AI chatbot. The interface is divided into three main sections: a left sidebar, a central chat area, and a right sidebar.

Left Sidebar:

- G-ONE** (Logo)
- CORE**
 - Dashboard
 - Plan Day
 - Chatbot (Selected)
- INTELLIGENCE**
 - Productivity Insights
 - History
 - Voice Assistant
- GOOGLE SERVICES**
 - Calendar
 - Tasks
 - Email
 - Contacts
 - Maps
- User Profile:** ARUN K... (Google Connected)

Central Chat Area:

- G-One Assistant** (Online) - Multi-agent AI - Real-time Google data
- Search bar:** Search for tasks, events, colleagues...
- Chat History:** 11:27 AM
- Message 1:** Hi! I'm G-One, your AI personal assistant. I can help with your **calendar, tasks, emails, maps**, and more. What would you like to know?
- Message 2:** summary last 5 important email
- Response:**
 - Recent Email Summary**
 - Based on your recent unread emails, here's a summary of the last 5 important emails:
 - * Remove distractions from photos instantly:** This email is from Adobe for Photographers, sent to you via mail@mail.adobe.com. The email likely contains information about a new feature or tool that can help remove distractions from photos instantly, which could be useful for photography enthusiasts or professionals. The subject line suggests that it might be a promotional email or a newsletter.
 - * Internship Application: Professional Development Opportunities at NYX Groups:** This email is from Gavi reddy Hruthik, sent from gavireddyhruthik@gmail.com. The subject line indicates that it's related to an internship application, and the email might contain information about professional development opportunities at NYX Groups. It's possible that the sender is reaching out to you about a potential internship or job opportunity.
 - * MEE6 mentioned you in A2D Channel:** This email is from Discord, sent via noreply@discord.com. The subject line suggests that someone (MEE6) has mentioned you in a Discord channel called A2D Channel. This email is likely a notification from Discord, and you might want to check the channel to see what the mention is about.
 - * 2.5m developers walk into a room:** This email is from Groq, sent from updates@groq.co. The subject line seems like a joke or a play on words, and the email might contain information about a developer-

Right Sidebar:

- Reasoning** (Icon)
- Clear** (Icon)
- Sync** (Icon)
- Actions** (Icon)
- Session** (Icon)
- HISTORY** (4)
- New Chat** (Icon)
- TODAY**
 - summary last 5 important email (2 messages)
 - Summarize my tasks (3 messages)
 - summarize last 5 email in short (3 messages)
- PREVIOUS 7 DAYS**
 - hello (3 messages)
- QUICK ACTIONS**
 - What's my next meeting?
 - Any urgent emails?
 - Summarize my pending tasks
 - Any conflicts today?
 - Plan my day
 - How's my travel looking?

PROPOSED WORK

VOICE ASSISTANT

The interface is titled "G-ONE" and features a sidebar with navigation options: CORE (Dashboard, Plan Day, Chatbot), INTELLIGENCE (Productivity Insights, History, Voice Assistant), and GOOGLE SERVICES (Calendar, Tasks, Email, Contacts, Maps). The main area displays a "Listening..." status with a microphone icon and a "Speak now" button. Below this, a search bar is present. The central panel shows a bar chart and a large red microphone icon. A notification bubble indicates "n summarize last 5 email in short" at 10:58 AM. The bottom section displays a list of voice sessions and quick commands. The voice sessions list includes: "summary last 5 important email" (3/17/2026), "summarize last 5 email in short" (3/17/2026), and "hello" (3/14/2026). The quick commands list includes: "What's my next meeting?", "Any new emails?", "Summarize my tasks", "Plan my day", "Check for conflicts today", and "What's on my agenda?". The settings section shows the language set to "us English (US)" and an "Auto-listen after response" toggle.

G-ONE

CORE

- Dashboard
- Plan Day
- Chatbot

INTELLIGENCE

- Productivity Insights
- History
- Voice Assistant

GOOGLE SERVICES

- Calendar
- Tasks
- Email
- Contacts
- Maps

Search for tasks, events, colleagues...

Listening... Speak now us English (US)

Press [Space] to toggle mic (continuous mode)

Input mode: Continuous

n summarize last 5 email in short 10:58 AM

n

Here's a brief summary of the last 5 emails:

- **Remove distractions from photos instantly:** An email from Adobe for Photographers about a feature to remove distractions from photos.
- **Internship Application: Professional Development Opportunities at NYX Groups:** An email from Gavi reddy Hruthik regarding an internship application at NYX Groups.
- **MEE6 mentioned you in A2D Channel:** A notification email from Discord mentioning that you were mentioned in a channel.
- **2.5m developers walk into a room:** An email from Groq with a subject line that seems to be a play on words, but the content is not specified.

Voice Sessions

- + New Session
- summary last 5 important email 3/17/2026
- summarize last 5 email in short 3/17/2026
- hello 3/14/2026

Quick Commands

- "What's my next meeting?"
- "Any new emails?"
- "Summarize my tasks"
- "Plan my day"
- "Check for conflicts today"
- "What's on my agenda?"

Settings

LANGUAGE

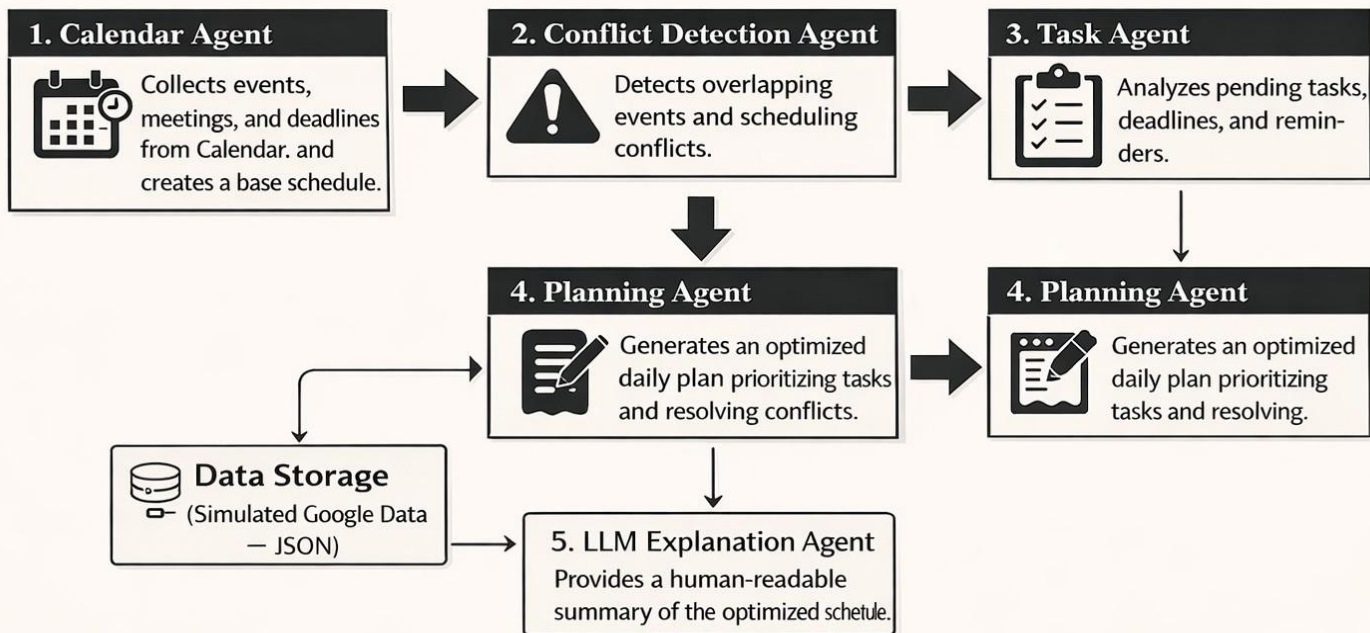
us English (US)

Auto-listen after response

APPLICATION EXAMPLE (POC)

Use Case: Automated Personal Task Scheduler

User Request: “Automatically organize my schedule and tasks for the day.”



REFERENCES

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THANK YOU